waves, which we currently think of as the stretching and squeezing of spacetime, could also be envisioned as the motion of gravitons expressing their wavelike nature.

Unfortunately, despite decades of hard work and extraordinarily intricate calculations, we haven't yet settled on a theory that's broadly accepted by the physics community. Not only have none of the ideas put forward been confirmed by particle experiments, it's not even clear that they *can* be. Ideally, we'd write down two theories and then work out that they make different predictions for experiments like those being carried out at the Large Hadron Collider. But this is a challenge when you're trying to distinguish between theories whose effects only become clear at energies many orders of magnitude higher than what LHC collisions can produce. This has led physicists to suggest solutions ranging from abstract arguments aimed at narrowing down the total range of possible universes to philosophical debates about how to make advances in areas of theory in which experimental evidence may never appear.

For those of us who hold out more hope for new data, our best bet for something that could give us hints about a theory of everything might well lie in cosmology—especially the study of the early universe. If you need data about particle interactions at extraordinarily high energies, finding new ways to examine the Big Bang is generally going to be easier than trying to build a particle collider the size of the Solar System.

We're already being nudged in this direction. So far, we've seen only a small handful of physical phenomena that we cannot explain within the Standard Model of particle physics (or very minor modifications thereof). The big ones, dark matter and dark energy, are strongly supported by observational evidence. But absolutely all of that evidence comes from cosmology and astrophysics. Figuring out what these mysterious cosmic components are and how they work might be the best hope we have to see where the theories should go next.

Another thing pointing us toward cosmology is the strange imbalance between matter and antimatter in the universe. Where our current theories suggest matter and antimatter should exist in equal quantities, our experience in the world and our ability to avoid constantly being annihilated by everything we touch shows us that regular matter is winning by a very wide margin. How that came about is still a mystery, but it's one for which